AI-Powered Regulatory Scrutiny, Forensic Document Intelligence, and Infrastructure Compliance Architecture for AICTE (PS 26261)

1. Multi-Modal Machine Learning Portfolio and Training Regimes

The transition of the All India Council for Technical Education (AICTE) from manual committee-driven scrutiny to an automated, evidence-linked regulatory intelligence framework requires a specialized portfolio of deep neural architectures[1]. Under the Approval Process Handbook (APH) 2024–2027, technical institutions submit complex multi-hundred-page dossiers spanning legal land deeds, municipal building plans, statutory No Objection Certificates (NOCs), audited financial statements, and faculty rosters[1]. Because this regulatory domain combines dense tabular data, bilingual administrative text, architectural CAD rasters, and high-incentive adversarial tampering, a single monolithic vision-language model is insufficient. Instead, the scrutiny engine must orchestrate five domain-specific subsystems: layout-aware Intelligent Document Processing (IDP), pixel-level forensic forgery localization, biometric seal and signature verification, panoptic architectural blueprint segmentation, and cross-institutional faculty deduplication[5].

A. Intelligent Document Processing and Layout-Aware OCR

Institutional filings exhibit extreme structural heterogeneity, ranging from digitally born PDF schedules exported from university ERP systems to low-DPI, skewed flatbed scans of notarized stamp-paper affidavits[3]. Selecting the optimal Intelligent Document Processing (IDP) backbone requires balancing spatial coordinate retention, tabular topological accuracy, and CPU/GPU inference throughput across high-volume scrutiny windows[7].

Architecture	Core Modality & Mechanism	Primary Strengths in AICTE Scrutiny	Operational Bottlenecks & Failure Modes	Architectural Role in PS 26261
LayoutLMv3	Multimodal Transformer; unified text and image patch masking	Superior sequence labeling (BIO tagging) and Key-Value	Requires an external upstream OCR engine to supply initial	Primary Semantic Entity Recognition (SER) and

	(16×16) with 2D spatial embeddings[10]	Extraction (KIE) on semi-structured legal affidavits and NOCs[11]	word bounding boxes and text tokens	Relation Extraction (RE) engine for legal deeds and certificates
Donut	OCR-free End-to-End Vision-Encoder Text-Decoder (Swin Transformer + BART decoder)	Bypasses OCR error propagation on degraded stamp papers; directly generates structured JSON trees	Prone to token hallucination on long multi-page tables; does not output explicit character-level bounding boxes	Fallback zero-shot document classifier and header metadata parser for degraded archival scans
Nougat	Visual Transformer trained for academic markup and LaTeX mathematical reconstruction	Accurately parses complex mathematical formulas, FSI/FAR equations, and curriculum syllabi	Suboptimal spatial localization for irregular multi-column administrative tables and municipal stamps	Specialized parser for academic curriculum annexation and fee-calculation annexures
PaddleOCR v4 / PP-OCRv5 + PP-StructureV3	DB++ text detection, SVTRv2 recognition, and SLANet-LCNet V2 table structure alignment[7]	Ultrafast inference; preserves exact cell coordinates (bbox) and HTML/DOM table topology (<thead>, rowspan, colspan)[7]	Requires downstream semantic alignment to link multi-page table continuations across PDF page breaks	Primary frontline OCR, layout segmentation, and high-density table extraction backbone[7]
TrOCR	Vision	High character	Computationall	Targeted

	Transformer (ViT/DeiT) encoder paired with an autoregressive RoBERTa/BART text decoder	recognition accuracy on isolated handwritten annotations, marginalia, and wet-ink dates	y expensive when applied to full-page dense documents; requires prior line-level cropping	line-level recognizer for handwritten Registrar remarks, notarized dates, and diary numbers

An analysis of the architectural trade-offs indicates that an OCR-free architecture such as Donut cannot serve as the primary scrutiny engine because AICTE's mandate requires strict visual explainability—every extracted data point must link back to an interactive bounding box on the source PDF canvas. Consequently, the optimal production pipeline couples **PaddleOCR (PP-OCRv5 / PP-StructureV3)** for frontline layout detection, text recognition, and table topology reconstruction with **LayoutLMv3-Large** for semantic entity linking and clause verification[7].

In the frontline stage, **PP-StructureV3** segments each page into distinct semantic zones—Text, Title, Table, Figure, Header, Footer, and Seal—using a PicoDet-LCNet_x1_0 backbone enhanced with Focal and Global Knowledge Distillation (FGD), which reduces model storage by 95% and achieves an average CPU inference time of 41 ms[12]. For text detection and recognition, PP-OCRv5 deploys a Differentiable Binarization (DB++) detector paired with a Scene Text Recognition (SVTRv2) network capable of unified multilingual recognition across English and Indian scripts[7]. When a tabular zone (such as an AICTE Faculty Roster containing PAN, EPF UAN, Qualification, and Date of Birth, or a Laboratory Equipment Invoice Register) is detected, the crop is routed to **SLANet (Structure Location Alignment Network)**[7].

Standard SLANet utilizes a lightweight PP-LCNet backbone with H-Swish activations[9]:

$$\text{H-Swish}(x) = x \cdot \frac{\text{ReLU6}(x+3)}{6}$$

However, vanilla SLANet exhibits limitations on complex multi-span regulatory tables due to fixed-length decoding steps and non-smooth gradient flow in deep layers[9]. To maximize table reconstruction fidelity, the architecture incorporates an improved SLANet configuration that replaces H-Swish in the feature fusion neck with the smooth, non-monotonic **Mish** activation function, incorporates an End-of-Sequence (**EOS**) dynamic termination mechanism to eliminate redundant decoding steps, and executes a multi-phase training strategy where input resolution scales to 512×512 during final convergence[9]. This configuration elevates Tree-Edit-Distance-based Similarity (**TEDS**) to 96.67%, accurately reconstructing merged cells (colspan, rowspan) and aligning cell coordinates with line-level OCR outputs to emit valid HTML and JSON table representations[9].

For unstructured and semi-structured legal documents—specifically AICTE APH

Format-1/Format-2 Affidavits, Fire Safety NOCs, Building Completion Certificates, and Land Registry Deeds—the word tokens and normalized 2D bounding boxes ($[x_0, y_0, x_1, y_1] \in [0, 1000]$) extracted by PP-OCRv5 are fed into **LayoutLMv3**[3].

LayoutLMv3 unifies text and visual modalities by tokenizing the document image into 16×16 linear patch embeddings alongside Byte-Pair Encoding (BPE) text tokens. Its pre-training regime optimizes a composite objective across three self-supervised tasks[10]:

$$\mathcal{L}_{\text{PreTrain}} = \mathcal{L}_{\text{MVLM}} + \mathcal{L}_{\text{MIM}} + \mathcal{L}_{\text{WPA}}$$

The first pre-training objective, Masked Visual-Language Modeling ($\mathcal{L}_{\text{MVLM}}$), randomly masks 30% of text tokens using span masking ($\text{Poisson}(\lambda = 3)$) while keeping their 2D layout coordinates intact, forcing the model to infer missing legal clauses from surrounding visual and spatial context[10]. Simultaneously, Masked Image Modeling ($\mathcal{L}_{\text{MIM}}$) masks 40% of image patches via blockwise masking and reconstructs discrete visual tokens generated by a dVAE codebook, enabling the model to learn structural typography, table gridlines, and watermark textures[11]. Complementing both is Word-Patch Alignment ($\mathcal{L}_{\text{WPA}}$, structurally analogous to Text-Image Alignment $\mathcal{L}_{\text{TIA}}$), a binary cross-entropy objective that predicts whether the corresponding image patch of an unmasked text token has been masked, enforcing granular cross-modal alignment between character glyphs and their underlying pixel patches[10].

During fine-tuning for AICTE Key Information Extraction (KIE), LayoutLMv3 is trained over a sequence length of $N = 512$ tokens (using a sliding window with a 128-token stride for multi-page deeds) using the **AdamW** optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$, weight decay $\lambda = 0.01$), a peak learning rate of 3×10^{-5} with a 10% linear warmup followed by cosine annealing decay, and a batch size of 32 across 50 epochs. A token-level classification head with a Conditional Random Field (CRF) layer optimizes the negative log-likelihood of BIO entity sequences, extracting critical regulatory entities (DEED_SURVEY_NO, PLOT_AREA_ACRES, NOC_EXPIRY_DATE, ISSUING_MUNICIPALITY, AFFIDAVIT_STAMP_SERIAL).

B. Document Forgery, Tampering, and Splicing Detection

In regulatory compliance scrutiny, adversarial manipulations rarely involve full-image generative synthesis; rather, they manifest as highly localized micro-forgeries where an applicant alters a single digit in an expiry date (e.g., changing 2024 to 2027 on a Fire Safety NOC), splices a valid municipal seal onto an unapproved blueprint, or copy-moves a signature block across multiple affidavits[17]. Because tampered pixels in document images routinely occupy well below 1% of the total canvas area, standard global image classifiers fail due to severe class imbalance[17].

Forensic Model / Technique	Underlying Signal & Architecture	Target Manipulation Vector	Strengths & Operational Characteristics	Critical Vulnerabilities & Mitigations
TruFor (Noiseprint++ + Transformer Fusion)	15-layer DnCNN self-supervised noise extractor + Cross-Modal RGB/Noise Transformer encoder + All-MLP anomaly & confidence decoders[6]	Splicing, digital compositing, generative inpainting (LaMa, SDXL), and out-of-camera editing histories[6]	Outputs pixel-level anomaly map, region-specific confidence map (suppresses false alarms), and global integrity score; resilient to WhatsApp/JPEG laundering[6]	Standard InstanceNorm2d at noise branch input destroys scalar noise statistics (AUC collapse from 0.747 to 0.554); requires ResAware BatchNorm + full-frame native extraction[21]
NFA-ViT (Noise-Guided Forgery Amplification)	Dual-branch Vision Transformer with Noise-Guided Amplification Attention (NAA) guided by Noiseprint++ masks[17]	Subtle, tiny-area localized text/digit edits and background ("stuff") region manipulations[17]	Amplifies weak manipulation cues across the image while leaving pristine regions unaffected[17]	Higher compute footprint; deployed as a secondary high-resolution magnifier on candidate text ROIs[17]
ManTra-Net	SRM filter bank + Wider VGG bayar-constrained feature extractor + 2D LSTM anomaly recurrence[17]	Classical copy-move and splicing boundaries[17]	Strong sensitivity to sharp local boundary transitions across 385 manipulation types	High false-positive rate on high-contrast binary text edges and scan compression artifacts[17]

ResNet-50 ELA & BAG Analysis	Multi-quality JPEG recompression differential (ΔQ) + 8×8 DCT Block Artifact Grid (BAG) alignment[19]	Post-JPEG splicing and localized patch saves in raster editors (Photoshop)[19]	Computationally fast; exposes 8×8 macroblock phase shifts and localized quantization residuals[19]	Degrades if the entire document is printed and re-scanned or heavily downsampled
CMFD (SIFT/ORB + Self-Attention)	Dense keypoint descriptor extraction + Transformer self-attention correlation matrix + RANSAC homography	Internal copy-move forgery (cloning a valid signature, stamp, or numeric digit from elsewhere on the same page)	Detects identical source-camera noise patches where Noiseprint++ sees uniform camera provenance	Struggles on uniform white background cloning; requires morphological filtering of keypoint clusters
Font Morphology CNN & PDF Stream Forensics	Character-level Siamese CNN (glyph stroke width, baseline, kerning) + PyMuPDF/Ghostscript XRef & font subset parser	Digit substitution (2021 $\rightarrow$ 2026) and PDF editing via Canva, Illustrator, or PDFtk	Catches vector-level PDF edits and font-embedding mismatches with 100% deterministic certainty	PDF stream forensics only applies to vector/hybrid PDFs; Font CNN handles rasterized scans

To achieve trustworthy, low-false-alarm forensic localization, the core neural engine integrates **TruFor** augmented with **Noise-Guided Forgery Amplification (NFA-ViT)** and a **ResAware-StatNoise** architectural modification[6]. TruFor operates on the principle that every imaging sensor (scanner or camera) and subsequent software processing pipeline imprints a microscopic, position-dependent residual trace across the image[6]. TruFor extracts this trace via **Noiseprint++**, a 15-layer Denoising Convolutional Neural Network (**DnCNN**) that transforms an input RGB image $x \in \mathbb{R}^{3 \times H \times W}$ into a full-resolution noise-sensitive fingerprint $r = R(x) \in \mathbb{R}^{1 \times H \times W}$ [19]. Unlike the original Noiseprint, which only modeled camera model

invariants, Noiseprint++ is trained via self-supervised contrastive learning to distinguish patches based on three simultaneous constraints: camera model, spatial coordinate (i, j) , and **editing history** (combinations of resizing, Gaussian blur, gamma correction, and JPEG recompression)[6].
During self-supervised pre-training on pristine high-resolution corpora from Flickr and DPReview, positive pairs consist of patches extracted from the exact same spatial grid coordinate of images sharing an identical camera and editing pipeline, while all other patches in the mini-batch serve as negatives[19]. The network minimizes the **InfoNCE contrastive loss**[6]:

$$\mathcal{L}_{\text{InfoNCE}} = -\sum_{i} \log \frac{\exp(\text{sim}(z_i, z_{i^+})/\tau)}{\sum_{k \neq i} \exp(\text{sim}(z_i, z_k)/\tau)}$$

where z_i denotes the L2-normalized embedding of patch i, $\text{sim}(u, v) = u^\top v$, and τ is the temperature hyperparameter[6].
A critical second-order architectural vulnerability emerges when deploying standard TruFor pipelines on localized document crops (such as isolated date blocks, faculty photographs, or stamp regions). Controlled seven-level diagnostic ablations reveal that while raw Noiseprint++ maps carry strong discriminative statistical signals—achieving an AUC of **0.747** when evaluated via a Multi-Layer Perceptron (MLP) over scalar noise statistics (Level L3)—the standard TruFor architecture places an InstanceNorm2d layer at the immediate input of the noise branch[21]. For a noise map $r \in \mathbb{R}^{C \times H \times W}$, InstanceNorm2d normalizes each channel independently using only that single sample's spatial mean μ_c and variance σ_c^2 [21]:

$$\hat{r}_{c,h,w} = \gamma_c \frac{r_{c,h,w} - \mu_c}{\sqrt{\sigma_c^2 + \epsilon}} + \beta_c$$

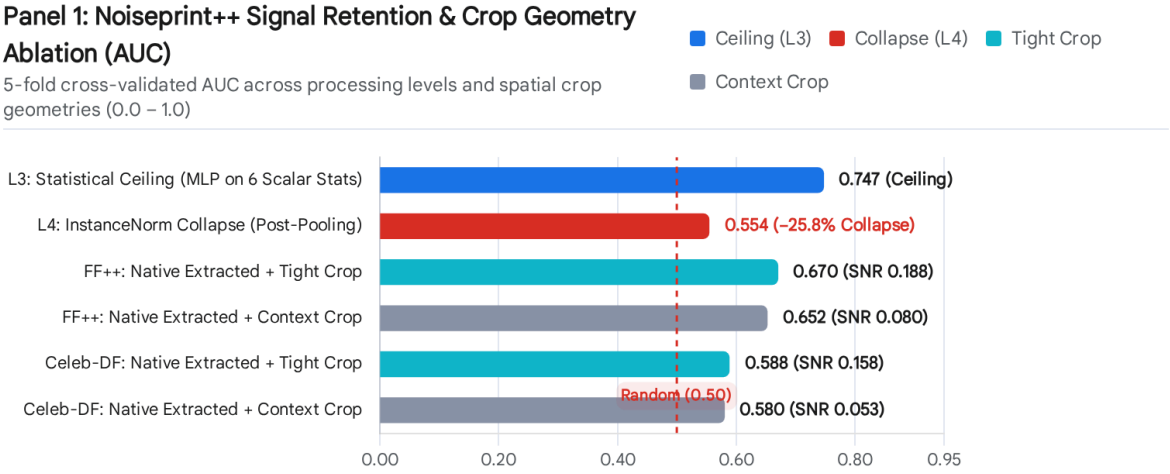
This spatial standardization silently annihilates the absolute magnitude and distributional variance of the sensor residual, causing the discriminative power of the noise statistics to collapse from **0.747** down to **0.554** (near random chance) at Level L4[21]. Furthermore, executing Noiseprint++ on small cropped regions degrades its receptive-field context[21].

AICTE DOCUMENT FORENSIC ARCHITECTURE & CV BENCHMARK

Eliminating **InstanceNorm Collapse** in Noiseprint++ Unlocks High-Precision Forensic Scrutiny

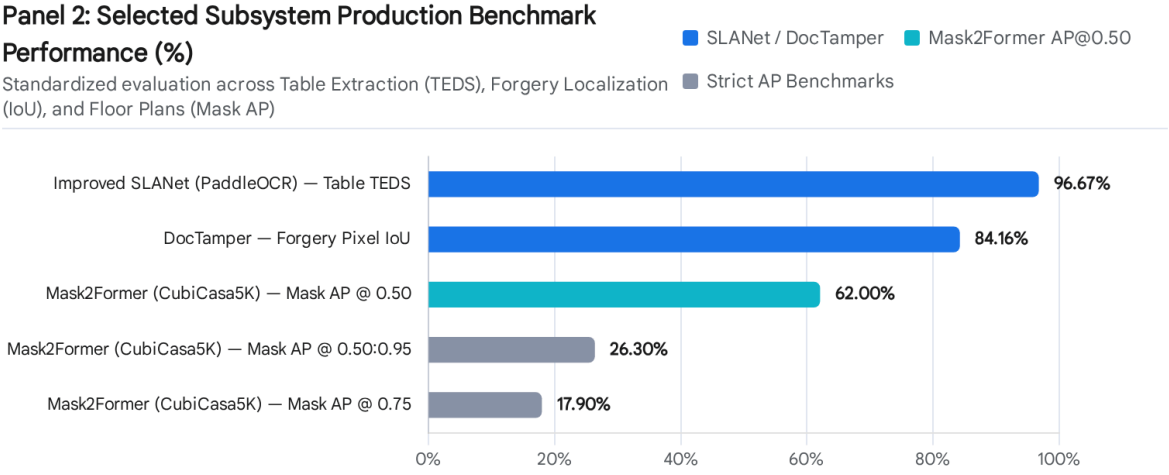
Panel 1: Noiseprint++ Signal Retention & Crop Geometry Ablation (AUC)

5-fold cross-validated AUC across processing levels and spatial crop geometries (0.0 – 1.0)



Panel 2: Selected Subsystem Production Benchmark Performance (%)

Standardized evaluation across Table Extraction (TEDS), Forgery Localization (IoU), and Floor Plans (Mask AP)



Top: Diagnostic ablation of Noiseprint++ residual maps demonstrates that standard InstanceNorm2d pooling (L4) destroys discriminative scalar noise statistics, dropping classification AUC from **0.747** (L3 ceiling) to **0.554**. Extracting Noiseprint++ on the full native frame prior to tight cropping preserves higher signal-to-noise ratio across both FF++ and Celeb-DF benchmarks. Bottom: Production benchmark performance of the selected subsystem models on their primary evaluation datasets.

Data sources: arXiv:2610.11683 (Noiseprint++ Forensics), MDPI Mathematics (Improved SLANet Table Recognition), PMC12190506 (DocTamper Benchmark), Hugging Face (Mask2Former CubiCasa5K)

To prevent this statistical collapse in the AICTE forensic pipeline, the architecture enforces three mandatory design rules[6]. First, under **Extract-Then-Crop Ordering**, Noiseprint++ is strictly computed on the full document page at native resolution prior to cropping any region of interest

(ROI), preserving the full receptive-field camera and scanner fingerprint[21]. When evaluating specific localized regions (such as a faculty face photo or date stamp), tight cropping following full-frame extraction yields a corpus Signal-to-Noise Ratio (SNR) of 0.188 (AUC 0.670) on FaceForensics++ and SNR 0.158 (AUC 0.588) on Celeb-DF, compared to SNR 0.080 (AUC 0.652) and SNR 0.053 (AUC 0.580) for wide context crops[21]. Second, within the cross-modal encoder, the noise branch replaces InstanceNorm2d with BatchNorm2d (**ResAware-Fusion**) and includes a parallel **StatNoise-Fusion** bypass branch that extracts 13 scalar noise statistics (mean, variance, skewness, kurtosis, and inter-quantile energy ratios) directly from the unnormalized Noiseprint++ map, fusing them with high-level visual features via a dedicated MLP[21]. Third, for fine-grained text and digit manipulations, the extracted noise trace r generates multi-scale noise-guided masks M_i^n that modulate the self-attention weights in the image branch via **Noise-Guided Amplification Attention (NAA)** inside **NFA-ViT**, amplifying subtle splicing discrepancies around character strokes without triggering false alarms on clean text boundaries[17].

The dual-decoder head of the forensic network is trained in two supervised phases following Noiseprint++ pre-training[6]. First, the pixel-level anomaly localization decoder $A \in [0,1]^{H \times W}$ is trained using a combination of class-weighted Binary Cross-Entropy ($\mathcal{L}_{\text{wBCE}}$) and **Dice Loss** ($\mathcal{L}_{\text{Dice}}$) to overcome the $< 1\%$ tampered-pixel sparsity[6]:

$$\mathcal{L}_{\text{Loc}} = \lambda_{\text{BCE}} \mathcal{L}_{\text{wBCE}}(A, Y) + \lambda_{\text{Dice}} \left(1 - \frac{2\sum_i A_i Y_i + \epsilon}{\sum_i A_i + \sum_i Y_i + \epsilon}\right.$$

Second, the confidence decoder $C \in [0,1]^{H \times W}$ and global integrity detector are trained by minimizing $\mathcal{L}_3 = \mathcal{L}_{\text{conf}} + \lambda_{\text{det}} \mathcal{L}_{\text{det}}$ [19]. The confidence map estimates region-specific uncertainty, down-weighting statistical anomalies in pristine areas (such as uniform white document margins or heavy scanner shadows) through confidence-weighted pooling, achieving balanced detection accuracy approaching 80% where fixed-threshold baselines degrade toward 50% random guessing[6]. On the 170,000-image **DocTamper** benchmark (spanning DocTamper-TestingSet, DocTamper-FCD, and DocTamper-SCD), this multimodal architecture achieves pixel-level localization exceeding 84.16% IoU at a 0.5 probability threshold[24].

To complement deep neural localization with deterministic mathematical proof for AICTE appellate proceedings, the forensic suite executes parallel structural checks across the raster and vector streams. Through Error Level Analysis (ELA) and Double JPEG Block Artifact Grid (BAG) inspection, the document image I is recompressed at a reference JPEG quality $Q = 90\%$ to produce I_{Q90}, yielding the amplified residual

$E(x,y) = |I(x,y) - I_{Q90}(x,y)| \times \gamma$ ($\gamma = 20$), while Discrete Cosine Transform (DCT) coefficient histograms are analyzed across 8×8 pixel blocks to detect phase misalignments in the JPEG grid that demarcate spliced patches from the pristine background grid[19]. Because cloning a valid signature or digit from the same page shares identical camera noise and JPEG quantization, Copy-Move Forgery Detection (CMFD) extracts dense SIFT and ORB keypoints alongside patch-level self-attention correlation matrices, applying agglomerative hierarchical clustering paired with RANSAC homography estimation to identify geometrically consistent translation and rotation vectors between cloned source and target regions. Simultaneously, using word and character bounding boxes from PP-OCRv5, a lightweight Siamese Character Morphology ConvNet computes stroke-width variance, baseline vertical offset (Δy), character slant angle (θ), and inter-character kerning differentials across adjacent characters in critical fields (Date, Area, Intake, Amount), where a localized Mahalanobis distance spike within a single word flags character substitution[7]. Finally, before rasterization, PyMuPDF (fitz) and pikepdf perform PDF Metadata and Ghostscript Stream Forensics by inspecting the raw cross-reference (xref) tables, incremental save trailers (%%EOF count), /Producer and /Creator metadata tags (flagging consumer editing tools like Canva, Adobe Photoshop, ILovePDF, or PDFtk), and embedded font subset prefixes to catch cases where a single modified line uses an unembedded ArialMT subset while the rest of the certificate uses a municipal TimesNewRomanPSMT subset.

C. Signature and Institutional Seal Verification

Every AICTE affidavit, building plan, and balance sheet requires authentication via wet-ink signatures and embossed or rubber-stamped seals of the Principal, Chairman/Secretary of the Trust/Society, Chartered Accountant, and Municipal Town Planner[3]. Localization is executed via a fine-tuned **YOLOv11m-Doc** object detector that scans full-page rasters at 1280×1280 resolution to localize four distinct classes: wet_signature, circular_seal, rectangular_designation_stamp, and notary_emblem. Training utilizes Complete IoU ($\mathcal{L}_{\text{CIoU}}$) for bounding box regression and Distribution Focal Loss ($\mathcal{L}_{\text{DFL}}$) to handle overlapping signatures superimposed directly over circular institutional stamps.
Once localized, detected stamp and signature crops undergo transformation into the HSV and CIE $L^*a^*b^*$ color spaces to isolate blue, violet, and red stamp ink from black printed background text via adaptive Otsu thresholding and morphological skeletonization. Circular seals are unwrapped into polar coordinates (r, θ) so that curved peripheral text can be read accurately by PP-OCRv5's dedicated seal text recognition module[14]. To verify whether a submitted signature or seal matches the institution's verified historical master repository—or to detect if a single digital PNG of a signature has been copy-pasted across dozens of pages—crops are embedded into a 512-dimensional hypersphere using a **Siamese ResNet-50** backbone trained with **Additive Angular Margin Loss (ArcFace)**:

$$\mathcal{L}_{\text{ArcFace}} = -\frac{1}{N}\sum_{i=1}^{N}\log\frac{e^{s(\cos(\theta_{y_i}+m))}}{e^{s(\cos(\theta_{y_i}+m))} + \sum_{j=1,j\neq y_i}^{n} e^{s\cos\theta_j}}$$

where θ_j is the angle between the feature vector x_i and class weight W_j, $s = 64.0$ is the hypersphere radius scaling factor, and $m = 0.50\ \text{rad}$ is the angular margin penalty. During scrutiny, a cosine similarity $\cos(\theta) < 0.72$ against the historical master record triggers a SIGNATURE_MISMATCH alert, whereas a pixel-exact structural similarity ($\text{SSIM} > 0.995$) between signatures across different documents triggers a DIGITAL_STAMP_CLONING forgery alert.

D. Architectural Floor Plan Segmentation and Area Computation

A primary vector of regulatory non-compliance involves submitting architectural blueprints where room dimensions are inflated in text labels or where circulation corridors are misclassified as instructional laboratory space to meet AICTE APH Chapter VII norms[4]. Automating this audit requires converting rasterized CAD blueprints into calibrated geometric polygons[28]. The floor plan parsing pipeline deploys **Mask2Former** initialized with a **Swin-Base** backbone (facebook/mask2former-swin-base-coco-instance) fine-tuned on the **CubiCasa5K** and **FloorPlanCAD** datasets[5]. On the CubiCasa5K validation benchmark, the baseline Swin-Base Mask2Former achieves a Mask $\text{AP@}[\text{IoU} = 0.50]$ of **0.620** ($\text{AP@}[0.50:0.95] = 0.263$ and $\text{AP@}0.75 = 0.179$)[5]. However, standard Mask2Former treats floor plan parsing as generic region segmentation, occasionally leaving breaks in continuous structural walls or misclassifying narrow hallways[30].

To enforce structural continuity, the pipeline integrates a parallel binary structural wall segmentation branch based on **RMBG-1.4** (phungpx/RMBG-1.4-wall-segmentation-cubicassa) alongside a Spatial-Geometric Boundary (**SGP**) recalibration module inspired by **PCP-Net**, which uses grouped convolutions ($P_{\text{group}} = \frac{k^2 C_{\text{in}} C_{\text{out}}}{G}$) and coordinate-aware feature recalibration to recover structurally challenging categories such as hallways, railings, and partitioned cubicles[30]. Furthermore, exploratory data analysis of CubiCasa5K reveals an extreme class imbalance ratio of **3,667×** in the training split (wall = 106,340 annotations vs. rare fixture classes = 29; validation imbalance ratio = 4,651× with 18,605 walls vs. 4 rare fixtures) alongside 10,175 tiny training instances and 1,773 validation instances with area $< 1,024\ \text{px}^2$ ($\approx 32 \times 32\ \text{px}$) across 4,223 unique image resolutions[28]. Training therefore standardizes resolution via aspect-ratio-preserving padding and combines multi-scale deformable attention with focal class weighting ($\gamma = 2.0$) and Dice mask loss[28].

Severe **Class Imbalance** and Micro-Instances in CubiCasa5K Mandate Specialized Blueprint Losses

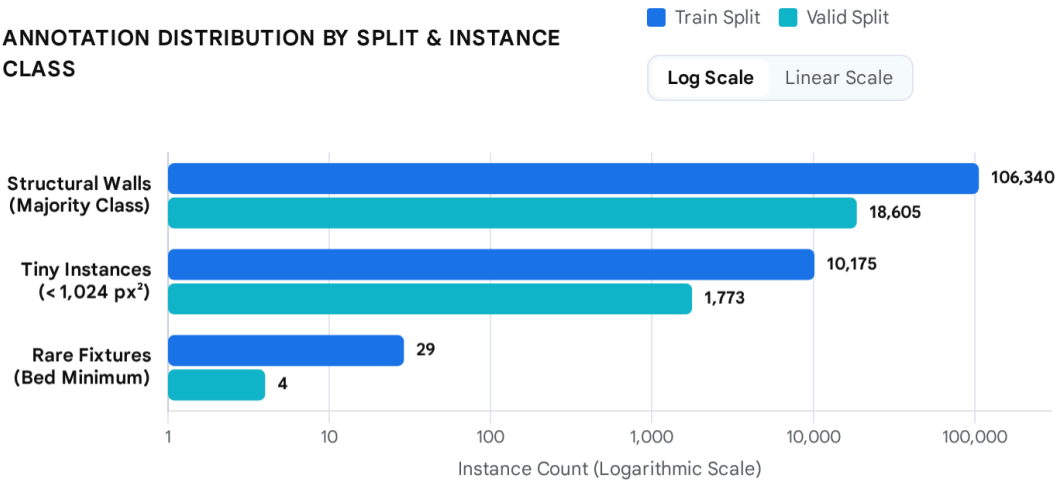
TOTAL ANNOTATIONS
270,465
4,976 blueprint images

BLUEPRINT SPLITS
4,228 / 748
Train (85%) · Valid (15%)

TRAIN IMBALANCE — Skew
3,667×
Wall 106k vs Bed 29

VALID IMBALANCE — Skew
4,651×
Wall 18.6k vs Bed 4

ANNOTATION DISTRIBUTION BY SPLIT & INSTANCE CLASS



Aspect-Ratio Preprocessing: 4,223 distinct image resolutions exist across the 4,228 training blueprints, requiring aspect-ratio-preserving padding prior to 640×640 / 1024×1024 batched training.

Exploratory data analysis of the CubiCasa5K architectural benchmark reveals **270,465** total polygon annotations across 4,228 training and 748 validation blueprints. Because structural wall segments outnumber rare fixture classes by **3,667×** in training (4,651× in validation) and 11,948 annotations occupy fewer than 1,024 square pixels, the AICTE blueprint parser pairs Mask2Former with focal class weighting and a dedicated RMBG-1.4 binary wall branch.

Data sources: Hugging Face (phungpx/cubicassa5k-coco), Hugging Face (phungpx/mask2former-swin-base-cubicassa)

Once Mask2Former and the RMBG-1.4 wall branch generate pixel masks for walls, doors, windows, stairs, and room polygons, the raster masks are vectorized into closed planar graphs using the **Ramer-Douglas-Peucker (RDP)** polygon simplification algorithm ($\epsilon = 2.0 \text{ px}$)[28].

To convert pixel area into physical square meters (m^2), a Hough Transform line detector paired with PP-OCRv5 locates graphic scale bars, dimension lines (e.g., 12.00 M), and printed room

dimension labels (e.g., CLASSROOM - 1: 9.0m x 7.5m)[7]. When a dimension line of length L_{px} pixels corresponds to parsed text L_{m} meters, the linear scale factor is $s = \frac{L_{\text{m}}}{L_{\text{px}}}\ (\text{m/px})$, allowing the physical carpet area of any segmented room polygon P with vertices $(x_1, y_1), \ldots, (x_n, y_n)$ to be computed via the **Shoelace Formula**:

$$\text{Area}_{\text{m}^2}(P) = s^2 \cdot \frac{1}{2} \left| \sum_{i=1}^{n} (x_i y_{i+1} - x_{i+1} y_i) \right|$$

The engine then performs a cross-modal discrepancy check by comparing the geometrically computed carpet area $\text{Area}_{\text{m}^2}(P)$ against the text area declared inside the room label on the blueprint and the tabular area claimed in the AICTE application portal. Any inflation exceeding 5% (such as a room text-labeled as 66.0 sq.m whose true geometric polygon measures 48.2 sq.m relative to the blueprint's own scale bar) triggers a BLUEPRINT_AREA_INFLATION deficiency flag, categorized across AICTE's four mandatory spatial buckets: **Instructional Area (lNA)**, **Administrative Area (ADA)**, **Amenities Area (AMA)**, and **Access and Circulation Area (ACA)**[27].

E. Campus Scrutiny Computer Vision and Ghost Faculty Deduplication

Physical infrastructure and human resource verification require cross-referencing visual evidence submitted in campus photographs and faculty profiles against national registries[1]. Institutions seeking Extension of Approval (EoA) or introduction of new courses must prove physical readiness of specialized laboratories and barrier-free PWD environments (ramps, lifts, tactile pathways, accessible toilets)[1]. Submitted geotagged photographs are evaluated by an ensemble of **RT-DETR-L** (Real-Time Detection Transformer) and **YOLOv8x** fine-tuned on 45 regulatory asset classes (cnc_lathe, universal_testing_machine, digital_oscilloscope, computer_workstation, wheelchair_ramp, fire_hydrant_extinguisher). To prevent institutions from submitting identical laboratory photographs across different departments or different colleges under the same Trust, every photograph is indexed using a 256-bit **Perceptual Hash (pHash / Wavelet wHash)** and a **DINOv2 ViT-B/14** visual embedding. Simultaneously, an EXIF parser extracts GPSLatitude, GPSLongitude, GPSAltitude, DateTimeOriginal, and camera model metadata from all uploaded site and laboratory photos. Using the Haversine distance formula and Shapely point-in-polygon geometric operations, the engine validates whether the photo's GPS coordinates fall inside the institution's declared cadastral land survey boundary and checks whether the campus plots adhere to the AICTE APH stipulation of at most **three plots** located within a maximum inter-plot distance of **2.0 km**, with all Academic, Instructional, Administrative, and Amenities areas housed on a single primary plot[4].

To satisfy mandatory Student-Faculty Ratios ($1:20$ for UG, $1:15$ for PG) and the $1:2:6$ cadre ratio without incurring full payroll costs, fraudulent institutions frequently register "Ghost Faculty"—individuals simultaneously listed on the rosters of multiple colleges[1]. The deduplication pipeline counters this through a three-tier biometric and cryptographic identity check. First, **RetinaFace** (ResNet-50 backbone) detects and aligns faculty portrait photographs via 5-point facial landmarks (expanding the bounding box by $1.3\times$ with square padding)[22], feeding the aligned 112×112 crop into **ArcFace (IResNet-100)** pre-trained on **Glint360k** to generate a 512-dimensional L2-normalized embedding $v \in \mathbb{R}^{512}$. Second, all 1.2M+ faculty embeddings across AICTE-approved institutions are indexed in **Qdrant / Milvus** using Hierarchical Navigable Small World (**HNSW**) graphs (M = 32, efConstruction = 200). When a new dossier is ingested, an Approximate Nearest Neighbor (ANN) cosine similarity search queries the national index in under $4\ \text{ms}$. Any cross-institutional match with $\cos(\theta) > 0.78$ where the PAN or Aadhaar number differs flags a BIOMETRIC_IDENTITY_FRAUD (the same individual registered under altered identity metadata), whereas a match on PAN, 12-digit EPFO UAN, or face embedding across two active institutions in the same academic year flags a CONCURRENT_GHOST_FACULTY violation[32].

2. Exhaustive Datasets Catalog and Synthetic Data Generation Pipeline

Training and benchmarking a regulatory scrutiny engine requires combining peer-reviewed academic datasets with a domain-specific synthetic data generator that replicates Indian statutory documents without exposing real institutional Personally Identifiable Information (PII).

A. Public and Benchmark Datasets Matrix

Subsystem Domain	Dataset Name	Scale & Composition	Ground-Truth Annotations & Format	Specific Utilization in PS 26261 Training
Document Layout & KIE	**PubLayNet** & **DocBank**	360k+ (PubLayNet) / 500k (DocBank) document pages[15]	COCO bounding boxes & polygon masks for Text, Title, Table, Figure, List[15]	Pre-training PP-StructureV3 (picodet_lcnet_x1_0_fgd_layout) for macro-layout segmentation[15]

Document Layout & KIE	**PubTabNet** & **TableBank**	568k HTML-annotated tables (PubTabNet)[15]	Cell bounding boxes + full structural HTML token sequences (<td>, colspan)[13]	Training SLANet-LCNetV2 for faculty roster and financial balance sheet table reconstruction[1][2]
Document Layout & KIE	**FUNSD**, **CORD**, **SROIE**, **RVL-CDIP**	199 noisy forms (FUNSD), 11k receipts (CORD), 400k docs (RVL-CDIP)	Token-level BIO entity tags, semantic key-value links, and 16 document classes	Fine-tuning LayoutLMv3 for KIE on semi-structured certificates and equipment tax invoices
Document Forgery & Tampering	**DocTamper (CVPR Benchmark)**	**170,000** bilingual tampered document images[24]	Pixel-accurate binary manipulation masks across TestingSet, FCD, and SCD splits[24]	Primary supervised training and cross-domain evaluation benchmark for TruFor and NFA-ViT[24]
Document Forgery & Tampering	**CASIA v1.0 / v2.0**, **Columbia**, **COVERAGE**, **CoMoFoD**, **NIST Nimble**	~25,000 composite spliced, copy-moved, and post-processed images	Binary ground-truth forgery masks + manipulation operation metadata	Pre-training the cross-modal RGB/Noiseprint++ anomaly decoder and benchmarking CMFD keypoint clustering
Signature & Stamp Forensics	**CEDAR**, **GPDS-Synthetic**, **SigComp 2011/2013**,	55k+ offline genuine/skilled-forged signatures;	Bounding boxes for seals/signatures + writer	Fine-tuning YOLOv11-Doc for seal/signature

	Tobacco800	1,290 complex business docs with seals	identity & genuine/forged labels	detection and Siamese ResNet-50 ArcFace for verification
Architectural Floor Plans	**CubiCasa5K (cubicassa5k-coco)**	**4,976** floor plans (4,228 train / 748 valid) with **270,465** annotations[28]	COCO polygon masks across 8 structural/fixture classes + over 80 room/icon subcategories[28]	Fine-tuning mask2former-swin-base-cubicassa and RMBG-1.4-wall-segmentation for room/wall parsing[5]
Architectural Floor Plans	**FloorPlanCAD**, **SESYD**, **CVC-FP**, **Rent3D (R3D)**	16,103 vector SVG CAD plans (FloorPlanCAD); 1,000 synthetic docs (SESYD)[29]	Vector primitive labels (35 classes) and structural room polygons[29]	Panoptic symbol spotting (doors, windows, fire exits) and cross-domain generalization testing[29]
Face Recognition & Deduplication	**Glint360k**, **VGGFace2**, **LFW**	17M images of 360k individuals (Glint360k); 3.31M images (VGGFace2)	Identity cluster labels and 5-point facial landmark coordinates	Training IResNet-100 ArcFace embeddings for cross-institutional Ghost Faculty deduplication

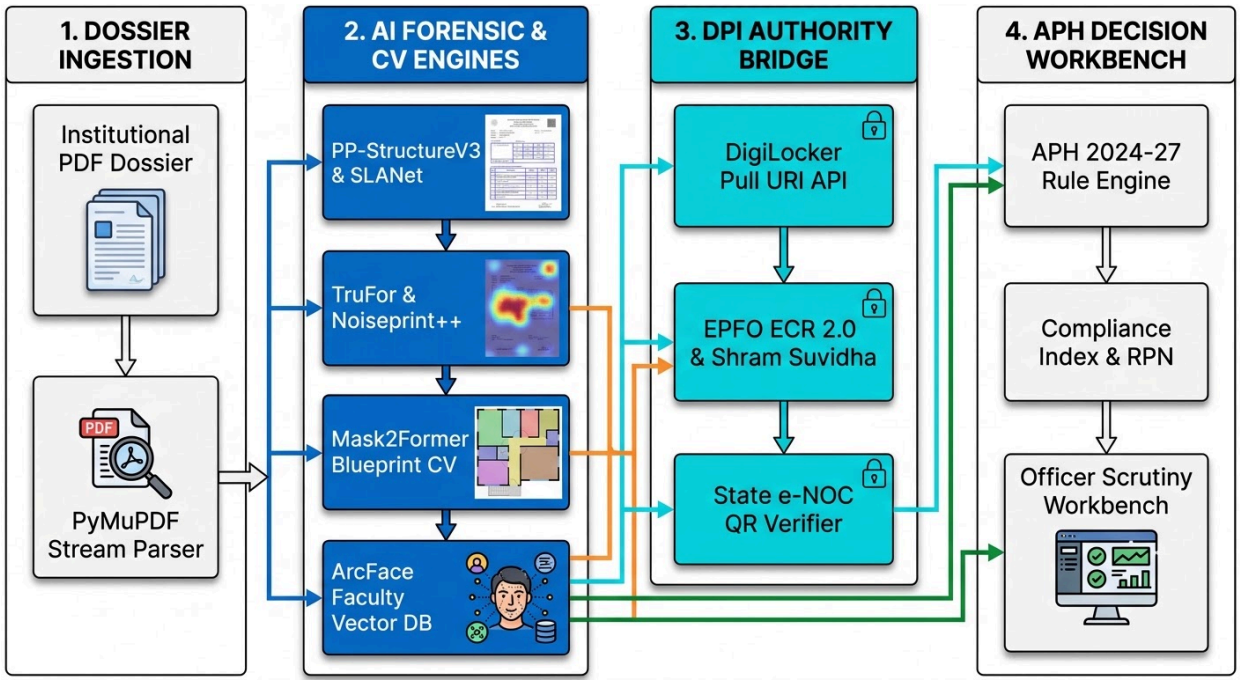
B. Algorithmic Pipeline for Synthetic AICTE Dossier Generation

Because real AICTE institutional filings contain confidential financial records, faculty PAN/Aadhaar numbers, and proprietary architectural drawings, training production models requires synthesizing a **10,000+ document AICTE Regulatory Corpus (AICTE-SynDoc-10K)** with pixel-exact ground-truth annotations for KIE, forgery masks, and geometric compliance labels. For synthetic affidavits following APH Format-1 and Format-2, Jinja2 and

WeasyPrint/LaTeX templates render authentic Rs. 100 Indian Non-Judicial e-Stamp headers (complete with synthetic 2D Aztec/QR barcodes, Unique Document Reference Numbers, and e-Stamp florals) followed by exact statutory text from AICTE APH 2024–27 Affidavit Formats[1]. The Faker library (en_IN and hi_IN locales) injects realistic Trust/Society names, AICTE Permanent IDs (1-4928174921), course intake tables, and Notary Public registration stamps, exporting exact character and word bounding boxes in FUNSD/COCO JSON format.
To generate doctored Fire Safety NOCs, PWD Certificates, and Land Deeds, an automated adversarial compositor introduces localized tampering operations across 40% of generated certificates while saving the exact binary ground-truth mask $Y \in \{0,1\}^{H \times W}$. This compositor executes date and survey number splicing by rendering a replacement year (such as 2027 over 2023) or land acreage (5.25 Acres over 2.10 Acres) using a slightly mismatched font family or weight (± 0.5 pt kerning offset, $\pm 1.2^{\circ}$ baseline tilt). It then applies localized JPEG re-compression patching by cropping the modified bounding box, compressing it at JPEG $Q_1 \in [60, 75]$ with a shifted $(dx, dy) \in [1, 7]$ pixel offset to break the 8×8 DCT grid, pasting it back into the master image compressed at $Q_2 \in [85, 95]$, and applying Poisson boundary blending[19]. Additionally, it performs stamp and signature cloning by extracting a genuine municipal stamp from one document, applying an affine rotation ($\theta \in [-5^{\circ}, 5^{\circ}]$) and alpha-channel multiplication, and superimposing it across multiple certificates.
For synthetic non-compliant architectural blueprints, procedural Python scripts (ezdxf paired with matplotlib and CairoSVG) generate multi-wing institutional floor plans containing Classrooms, Laboratories, Drawing Halls, Principal Offices, Faculty Cabins, Corridors, and Toilets[27]. To train the cross-modal discrepancy detector, the generator deliberately introduces regulatory violations in 50% of blueprints, such as drawing a Chemistry Lab polygon whose true geometric coordinate area is $44.0\ \mathrm{m}^2$ (violating the $66.0\ \mathrm{m}^2$ AICTE norm) while printing the false text label CHEMISTRY LAB - 66.0 SQ.M inside the polygon, or omitting mandatory PWD wheelchair ramps and secondary fire exit staircases[27]. Finally, to simulate physical submission workflows and adversarial laundering (such as WhatsApp compression or printing and re-scanning to hide digital splices)[6], all synthetic documents pass through a stochastic Albumentations pipeline incorporating Perspective(scale=(0.02, 0.06)), ElasticTransform (simulating page curl near binder spines), GaussNoise(var_limit=(10.0, 45.0)), spatially varying illumination gradients, halftoning/dithering, and multi-stage ImageCompression(quality_lower=55, quality_upper=90).

3. External Issuing-Authority Integration Architecture

Visual and forensic scrutiny alone cannot detect a document that is freshly printed with pristine typography but contains entirely fictitious certificate numbers. The platform therefore implements a **National Digital Public Infrastructure (DPI) Authority Bridge** to cryptographically verify documents against their primary issuing repositories[33].



A. DigiLocker, NAD, and API Setu Verification Flow

To verify Affiliating University Consent Letters, Land Registry Deeds, Fire NOCs, and Faculty Degree Certificates issued via **DigiLocker / EntityLocker** and the **National Academic Depository (NAD)**, the platform registers as a Requester Partner on **API Setu**[37]. During dossier submission, the institutional representative authenticates via DigiLocker's OAuth 2.0 endpoint (/oauth2/1/token) using Proof Key for Code Exchange (**PKCE**)[38]. For every authorization request, the AICTE bridge dynamically generates a high-entropy cryptographic code_verifier (43–128 characters) and derives the code_challenge via SHA-256 hashing and Base64URL encoding, exchanging the authorization code alongside client_id, client_secret, grant_type="authorization_code", redirect_uri, and code_verifier[38].

All documents issued in compliance with DigiLocker Technical Specifications (DLTS) adhere to the canonical URI structure IssuerId-DocType-DocId (for example, in.gov.maha.maharerait-BLDGCOMP-202689412), where IssuerId is a mandatory unique pure-alpha case-insensitive string identifying the state department or university[37]. When verifying an uploaded certificate, the AICTE Authority Bridge invokes the REST-based **Pull URI Request API** (and Pull Document Request API) over HTTPS POST, passing the 36-character DigiLockerId, DocType, and search criteria parameters[37]. Every request header carries an x-digilocker-hmac authentication digest computed over the request payload and timestamp (ts), where ts must fall strictly within 30 minutes of the current server clock to prevent replay attacks[37]:

$$\text{x-digilocker-hmac} = \text{HMAC-SHA256}(\text{client_secret}, \text{client}$$

The Issuer repository responds synchronously with a Pull URI Response XML containing DocDetails (name, dob in DD-MM-YYYY format, phone, URI), DocContent (Base64 byte-encoded official PDF file contents when the format attribute is requested as "pdf" or "both"), and DataContent (Base64 byte-encoded structured certificate metadata in XML format)[37]. The AICTE engine decodes DataContent and performs a deterministic field-by-field reconciliation against the values extracted by LayoutLMv3 from the applicant's uploaded PDF[11]. Even if the applicant's PDF passes visual inspection, any discrepancy between the uploaded text and the authoritative DataContent XML (such as an altered expiry date or sanctioned intake count) immediately flags MISMATCH_DETECTED[37].

B. Cryptographic QR Verification, EPFO Shram Suvidha, and MCA21 Handshakes

State Fire Directorates, Pollution Control Boards, and Municipal Development Authorities embed high-density 2D QR codes or digitally signed JSON Web Signatures (JWS) on e-District NOCs. The scrutiny pipeline decodes QR payloads using pyzbar and OpenCV WeChatQRCode (which employs a super-resolution CNN to recover blurred or low-DPI QR matrices). For cryptographically signed payloads following UIDAI and e-District PKI standards, the verifier extracts the compressed payload and verifies the **ECDSA (secp256r1)** or **RSA-2048 SHA-256** digital signature against the AICTE local trust store of State Certifying Authority X.509 public keys. For URL-based verification QR codes, the URL parser inspects the Fully Qualified Domain Name (FQDN) against a strict whitelist of official NIC and State portals (*.gov.in, *.nic.in), blocking adversarial attacks where fraudulent applicants generate a custom QR code pointing to a lookalike spoofing domain hosting a fabricated verification page.

To eliminate "Ghost Faculty" and "Paper Appointments" (where faculty members lend their degree certificates to an institution without actually working or drawing a salary there), the platform integrates with the **Employees' Provident Fund Organisation (EPFO)** via the **Unified Shram Suvidha Portal (USSP)** and Aggregator APIs under the **Social Security Code, 2020** and the **EPF Scheme, 2026**[32]. Any technical institution employing 20 or more persons is statutorily required within 30 days to register on the Unified Shram Suvidha Portal using its corporate PAN and Class 3 Digital Signature Certificate (DSC) to obtain a unique Establishment PF Code[32]. Every faculty member submitted in the AICTE roster must possess a 12-digit **Universal Account Number (UAN)** seeded with Aadhaar and PAN, activated via Aadhaar-based Face Authentication (FAT) through the UMANG/EPFO architecture[32].

Under Paragraphs 20(1) and 28(3) of the EPF Scheme, 2026, employers must file the monthly **Electronic Challan cum Return (ECR 2.0)** and remit 12% employer (split as 8.33% to EPS and 3.67% to EPF) and 12% employee contributions within 15 days of the close of every month, with real-time interest (@ 12% p.a.) and Section 14B damages (5% to 25%) assessed for even a 24-hour delay[32]. Crucially, under the 2026 enforcement of the redefined wage rules, basic wages used for PF calculation cannot be less than **50% of gross remuneration**—where excluded allowances exceed half of total remuneration, the excess is statutorily deemed to be wages and added back to the contribution base[42]. The AICTE Authority Bridge cross-verifies whether the faculty member's UAN shows active monthly ECR 2.0 remittances under the

applicant institution's Establishment Code for the preceding academic months, whether the same UAN has active overlapping ECR remittances from another establishment code without a recorded Date of Exit (Mark Exit in the e-SEWA portal), and whether the remitted PF contribution matches the salary scale claimed in the institution's AICTE Aquittance Roll under the 50% wage add-back rule[42]. Complementing payroll checks, for institutions established by a Section 8 Company, Society, or Charitable Trust, the Authority Bridge queries the **Ministry of Corporate Affairs (MCA21 v3)** API and NITI Aayog NGO Darpan registry using the Corporate Identity Number (CIN), Trust Registration Number, and establishment PAN to verify active legal standing and absence of blacklisting[3].

To ensure the system functions strictly as a transparent decision-support engine rather than an opaque black box, every external verification call resolves into one of six finite state machine categories:

Verification State	Trigger Condition	Automated Scrutiny Impact & Officer Action
VERIFIED	Cryptographic hash, URI payload, and extracted fields match the issuing authority's response within tolerance[37].	Green badge; zero risk penalty; auto-clears checklist item.
MISMATCH_DETECTED	Authority confirms the URI/Certificate ID exists, but key attributes (e.g., Land Area, Expiry Date, UAN Employer Code) contradict the uploaded PDF[37].	Critical Red Flag; triggers fatal forgery/tampering alert and populates evidence diff table.
EXPIRED	Document is authentic in the authority registry, but the statutory validity window has lapsed prior to the Academic Year 2026–27 cutoff.	Amber/Red compliance flag; auto-drafts a rectifiable deficiency notice requiring renewal submission.
VERIFICATION_PENDING	Asynchronous webhook or multi-step OAuth consent from the institution's nodal officer is awaiting	Blue status pill; holds final EoA generation until handshake resolves or times out.

	completion.	
SOURCE_UNAVAILABLE	Upstream state portal or API Setu endpoint returns HTTP 502/503/504 after exponential backoff retries ($2s, 4s, 8s$).	Neutral Slate badge; prevents false penalization of the institution and routes document to manual scrutiny queue.
MANUAL_REVIEW_REQUIRED	Legacy offline deed (e.g., pre-digitization 1985 physical land registry) or OCR confidence $<$ prevents automated URI construction.	Amber review badge; highlights specific bounding boxes for Regional Officer visual verification.

4. AICTE Approval Process Handbook Rule Engine and Compliance Scoring

A. Deterministic Mathematical Formulations of APH Norms

The AICTE Approval Process Handbook (APH) 2024–2027 introduces fundamental regulatory reforms designed to align technical education with the National Education Policy (NEP) 2020, including multi-disciplinary flexibility, working-professional skilling provisions, bringing Undergraduate Computer Applications (BCA) and Management (BBA) courses in General Degree Colleges under the AICTE umbrella, and granting Extension of Approval (EoA) for up to 3 years for outstanding compliant institutions[1]. Most significantly for infrastructural scrutiny, APH 2024–2027 replaces rigid legacy acreage tables with dynamic, location-specific **Floor Space Index (FSI) / Floor Area Ratio (FAR)** computations across Mega, Metro, Urban, and Rural classifications[1].

Let an institution operate P approved programmes, where each programme $p \in \{1, \ldots, P\}$ has an approved annual student intake I_p and a course duration D_p (in years)[4]. Under Appendix 4 and Chapter VII of the APH, the Total Required Built-up Area (TB_{req} , in m^2) is calculated for the entire duration of the course as the sum of four mandatory spatial components[4]:

$$TB_{\text{req}} = (\text{INA}_{\text{req}} + \text{ADA}_{\text{req}} + \text{AMA}_{\text{req}}) + \text{ACA}_{\text{req}}$$

Within this formulation, INA_{req} represents the **Instructional Carpet Area** (Classrooms at

$1.0–1.1\ \mathrm{m}^2$ per student, Tutorial Rooms at 25% of classrooms, Laboratories at $66\ \mathrm{m}^2$ per lab module, Workshops, Drawing Halls, Computer Centers, and Library), with the explicit stipulation that in the case of allied branches in Engineering and Technology, a maximum of 30% **of Laboratories may be shared** ($\mathrm{Lab}_{\mathrm{req}}^{\mathrm{allied}} = 0.70 \times \mathrm{Lab}_{\mathrm{nominal}}$)[4]. The term $\mathrm{ADA}_{\mathrm{req}}$ denotes the **Administrative Carpet Area** (Principal Office, Board Room, Office All Inclusive, Department Offices, Cabins for Head of Department and Faculty), where for Post Graduate programmes, the Administrative Area of Under Graduate programmes may be shared[27]. The term $\mathrm{AMA}_{\mathrm{req}}$ captures the **Amenities Carpet Area** (Barrier-free Toilets for Boys, Girls, and PWD, Boys and Girls Common Rooms, Cafeteria, First-Aid cum Sick Room, and Stationery Store), while $\mathrm{ACA}_{\mathrm{req}}$ represents the **Access and Circulation Area** (Corridors, Staircases, Entrance Lobbies, and structural wall footprints), mandated at 25% of the total carpet area such that $\mathrm{ACA}_{\mathrm{req}} = 0.25 \times (\mathrm{INA}_{\mathrm{req}} + \mathrm{ADA}_{\mathrm{req}} + \mathrm{AMA}_{\mathrm{req}})$[27].

Given the prevailing municipal **FSI / FAR** norm Φ_{loc} declared by the Competent Development Authority for the institution's specific location (Mega, Metro, Urban, or Rural), the minimum required land area L_{req} (in m^2, where $1\ \mathrm{Acre} = 4046.86\ \mathrm{m}^2$) is governed by[2]:

$$L_{\mathrm{req}} = \max\left(\frac{TB_{\mathrm{req}}}{\Phi_{\mathrm{loc}}},\ L_{\mathrm{legacy_floor}}\left(\mathrm{Zone}, \mathrm{Year}_{\mathrm{est}}\right)\right)$$

where $L_{\mathrm{legacy_floor}}$ accounts for Annexure 12 historical norms (such as $2.5\ \mathrm{Acres}$ Urban vs. $7.5\ \mathrm{Acres}$ Rural for legacy Engineering campuses, while permitting institutions to start other Technical or Non-Technical educational courses in surplus land or built-up area arising out of reduced norms)[3]. Furthermore, the geometric spatial validator enforces the **Contiguity & Multi-Plot Constraint**, which mandates that required land shall be in a maximum of **three plots** ($N_{\mathrm{plots}} \leq 3$), the inter-plot Haversine distance shall not exceed **2.0 km** ($\max_{i,j} d_{\mathrm{Haversine}}\left(\mathrm{Plot}_i, \mathrm{Plot}_j\right) \leq 2.0\ \mathrm{km}$), and the entire Academic, Instructional, Administrative, and Amenities area must reside on a single primary plot ($(\mathrm{INA} \cup \mathrm{ADA} \cup \mathrm{AMA}) \subseteq \mathrm{Plot}_1$), leaving remaining plots exclusively for sporting infrastructure, hostels, and staff accommodation[4].

For human resource scrutiny, let $S_{\mathrm{UG}} = \sum_{p \in \mathrm{UG}} I_p \cdot D_p$ and $S_{\mathrm{PG}} = \sum_{q \in \mathrm{PG}} I_q \cdot D_q$ denote the total sanctioned student strength across Under Graduate and Post Graduate programmes. The minimum required regular faculty strength F_{req} (after excluding any faculty

flagged as CONCURRENT_GHOST_FACULTY or lacking verified EPFO ECR 2.0 remittances) is:

$$F_{\text{req}} = \left\lceil \frac{S_{\text{UG}}}{20} \right\rceil + \left\lceil \frac{S_{\text{PG}}}{15} \right\rceil$$

Following the $1:2:6$ Cadre Ratio (Professor : Associate Professor : Assistant Professor), the required distribution across cadres is:

$$F_{\text{Prof}} = \left\lfloor \frac{1}{9} F_{\text{req}} \right\rfloor, \quad F_{\text{Assoc}} = \left\lfloor \frac{2}{9} F_{\text{req}} \right\rfloor, \quad F_{\text{Asst}} = F_{\text{req}} - (F$$

Up to 10% of required faculty positions may be fulfilled by qualified Adjunct Faculty or Industry Resource Persons in accordance with APH Annexure 9 guidelines[3].

B. Institutional Compliance Index and Risk Priority Number Algorithms

To synthesize thousands of granular checks into an actionable decision-support metric for AICTE Regional Officers and Expert Visit Committees (EVC)[1], the engine computes two deterministic indices. The **Institutional Compliance Index (**$\text{ICI} \in [0, 100]$ **)** aggregates five weighted domain sub-scores—Infrastructure & Built-up Area (S_{infra}, weight $w_1 = 0.25$), Faculty SFR & Cadre Compliance (S_{fac}, $w_2 = 0.25$), Statutory NOC & Legal Validity (S_{legal}, $w_3 = 0.20$), Authority Bridge Reconciliation (S_{auth}, $w_4 = 0.15$), and Financial & Fee Compliance (S_{fin}, $w_5 = 0.15$)—modulated by a non-linear **Forensic Integrity Gate (**Γ_{forensic}**)**:

$$\text{ICI} = \Gamma_{\text{forensic}} \cdot \sum_{k=1}^{5} w_k S_k, \quad \text{where } \Gamma_{\text{forensic}} = \prod_{m \in \mathcal{M}_{\text{fatal}}} (1 - \mathbb{I}($$

If a **fatal fraud** event is confirmed with high TruFor anomaly probability ($P_{\text{tamper}} \geq 0.85$) and confidence ($C_{\text{conf}} \geq 0.80$)—such as a doctored Land Deed, a forged Fire Safety NOC,

or confirmed biometric Ghost Faculty—the gate Γ_{forensic} caps the effective compliance status and routes the dossier directly to STANDING_HEARING_COMMITTEE_SHOW_CAUSE regardless of nominal infrastructural scores[6]. Conversely, **rectifiable deficiencies** (such as a 4% classroom carpet area shortfall or a Fire NOC renewal application pending municipal sign-off) proportionally reduce S_k without triggering the fatal fraud gate, routing the dossier to DEFICIENCY_RECTIFICATION_WINDOW.

Every detected anomaly a_i is simultaneously ranked in the Scrutiny Workbench using a continuous forensic **Risk Priority Number (**$\text{RPN}(a_i) \in [1, 1000]$**)**:

$$\text{RPN}(a_i) = \text{Severity}(a_i) \times \text{Occurrence}(a_i) \times \text{ForensicCert}$$

Within this triage product, $\text{Severity}(a_i) \in [1, 10]$ measures statutory impact weight (10 for forged land title/NOC or ghost faculty; 7 for instructional lab area deficit $> 15\%$; 4 for missing Associate Professor cadre; 2 for minor OCR formatting variance), $\text{Occurrence}(a_i) \in [1, 10]$ quantifies systemic extent across the filing (10 when repeated across > 5 documents or cross-institutional matches in Qdrant; 5 when isolated to a single annexure table; 1 for a single cell), and $\text{ForensicCertainty}(a_i) \in [1, 10]$ reflects calibrated AI and cryptographic confidence (10 for deterministic API Setu MISMATCH_DETECTED or PDF stream font-subset injection; $8–9$ for TruFor Noiseprint++ pixel anomalies with $C_{\text{conf}} > 0.85$; 5 for uncalibrated blueprint OCR scale warnings)[6].

5. Master Prototype Build Specification for Autonomous Execution

The following section contains the complete, self-contained **Master Prototype Build Prompt** engineered for direct execution by the autonomous coding agent (**Antigravity**). It translates every architectural, forensic, regulatory, and SIH26246 UI/UX mandate into concrete code specifications, module mappings, CSS token systems, and mock data structures.

ANTIGRAVITY AUTONOMOUS EXECUTION PROMPT: AICTE REGULATORY SCRUTINY & FORENSIC

INTELLIGENCE ENGINE (PS 26261)

1. SYSTEM IDENTITY & PRODUCT MANDATE

Build a complete, production-grade, interactive web application for **Smart India Hackathon Problem Statement ID 26261**: **"AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification"** for the **All India Council for Technical Education (AICTE), Ministry of Education, Government of India (Regulation Bureau)**.

Core Product Philosophy:

- This platform is an **Evidence-Linked AI Decision-Support & Forensic Scrutiny Workstation** for AICTE Regional Officers, Scrutiny Committees, and Expert Visit Committees (EVC)—NEVER an opaque automated black-box approval tool.
- Every anomaly flagged by AI MUST display interactive visual bounding boxes, pixel-level Error Level Analysis (ELA) / Noiseprint++ heatmaps, exact AICTE Approval Process Handbook (APH 2024–27) rule citations, and deterministic Authority Bridge verification badges.
- Strictly separate **Fatal Fraud** (forged Fire NOC, doctored Land Deed survey acreage, biometric/UAN Ghost Faculty across colleges) from **Rectifiable Deficiencies** (pending municipal NOC renewal, minor 3.8% tutorial room area shortfall, missing notary page).

2. MANDATORY UI/UX DESIGN SYSTEM: REPLICATING SIH26246 ("LIQUID GLASS CIVIC & REGULATORY INTELLIGENCE UI")

2.1 Visual Formula & Atmospheric Canvas

- **Aesthetic Identity:** Apple Vision Pro spatial glass × High-Density Government Regulatory Workstation × Modern Indian Digital Public Infrastructure (DPI). NEVER use generic dark-neon cyberpunk AI aesthetics.
- **Light Mode Canvas (default):** Base background #F4F7FB layered with atmospheric radial mesh depth:background-image: radial-gradient(ellipse 80% 50% at 50% -10%, rgba(14, 165, 233, 0.06), transparent 70%), radial-gradient(ellipse 60% 40% at 85% 20%, rgba(67, 56, 202, 0.04), transparent 60%);
- **Dark Mode Canvas:** Base background #0B132B with subtle cyan/indigo atmospheric radial glow (rgba(14, 165, 233, 0.09) and rgba(67, 56, 202, 0.08)).
- **Liquid Glass Card Token (.liquid-glass-card):**
  - Light: background: rgba(255, 255, 255, 0.75); backdrop-filter: blur(24px) saturate(150%); border: 1px solid rgba(226, 232, 240, 0.8); box-shadow: inset 0 1px 0 0 rgba(255, 255, 255, 0.95), 0 10px 30px -10px rgba(15, 23, 42, 0.06);
  - Dark: background: rgba(15, 23, 42, 0.75); backdrop-filter: blur(24px) saturate(150%); border: 1px solid rgba(30, 41, 59, 0.8); box-shadow: inset 0 1px 0 0 rgba(255, 255, 255, 0.08), 0 10px 30px -10px rgba(0, 0, 0, 0.4);

2.2 Typography & 100% Bilingual Localization (EN | हिन्दी)

- Load Google Fonts in app/layout.tsx: Instrument Serif, Plus Jakarta Sans, JetBrains Mono, Noto Sans Devanagari, and Khand.
- **Editorial Hero Headings:** Use Instrument Serif italic accents inside crisp sans headings (e.g., Institutional Regulatory Scrutiny Engine).
- **Body & UI Controls:** Plus Jakarta Sans (--font-sans).
- **Regulatory Codes, Hashes, UANs & Coordinates:** JetBrains Mono (--font-mono).
- **KPI Numerals:** Always enforce tabular-nums with fluid scaling clamp(1.75rem, 2.4vw, 2.125rem).
- **Full Bilingual Toggle (EN | हिन्दी):** Implement a React Context (LanguageContext) storing 'en' | 'hi'. When switched to 'hi', render 100% of navigation items, KPI headers, alert banners, APH rule citations, and deficiency notices in authentic formal regulatory Hindi (Noto Sans Devanagari / Khand).

2.3 The Unified 6-Column Metric Grid (SIH26246 Signature Component)

Render at the top of /overview using:grid grid-cols-1 sm:grid-cols-2 lg:grid-cols-3 xl:grid-cols-6 divide-y sm:divide-y-0 sm:divide-x divide-slate-200/80 dark:divide-slate-800 liquid-glass-card rounded-2xl overflow-hidden Each of the 6 cells MUST contain:

1. **2px Top Severity Rule:** Top border bar colored by state (#DC2626 severe/fraud, #D97706 moderate deficiency, #059669 verified compliant, #4338CA national registry archive).
2. **Category Label:** text-xs font-semibold uppercase tracking-[0.06em] text-slate-500 dark:text-slate-400 line-clamp-2 min-h-[2rem].
3. **Locked Baseline Row:** Flex row (flex items-baseline justify-between gap-2 my-1.5) where the large tabular-nums metric sits on the **exact same vertical baseline** as a compact trend pill (px-2 py-0.5 rounded-full border text-[11px] font-bold inline-flex items-center gap-1) with Lucide ArrowUpRight, ArrowDownRight, or Minus.
4. **Subtitle Row:** text-xs text-slate-500 dark:text-slate-400 truncate.
5. **Micro-interaction:** hover:-translate-y-0.5 transition-all duration-200 hover:bg-white/40 dark:hover:bg-slate-800/40.

The 6 exact KPI cards on /overview:

1. **Total Applications Ingested:** 4,820 | +12.4% (#4338CA Indigo rule) | AY 2026-27 Scrutiny Cycle
2. **Cleared via Automated Pre-Scrutiny:** 3,194 | 66.2% (#059669 Emerald rule) | ICI Score ≥ 90 & Zero Fatal Flags
3. **Rectifiable Deficiencies Issued:** 1,182 | -4.1% (#D97706 Amber rule) | 15-Day EVC Rectification Window
4. **Fatal Forgery & Tampering Alerts:** 148 | +18.9% (#DC2626 Red rule) | TruFor Noiseprint++ Confidence ≥ 85%
5. **Cross-Campus Ghost Faculty Flagged:** 296 | 94 Colleges (#DC2626 Red rule) | ArcFace 512-D & EPFO UAN Match

6. **Authority Bridge Live Handshakes:** 98.4% | 142ms avg (#059669 Emerald rule) | DigiLocker Pull URI • EPFO ECR 2.0

2.4 Priority Intervention / High-Risk Alert Banner (SIH26246 Pattern)

Directly below the 6-column KPI grid, render an interactive glass banner (glass-interactive border border-amber-500/30 bg-amber-500/5 dark:bg-amber-500/10 rounded-2xl p-4 sm:p-5 flex flex-col lg:flex-row items-start lg:items-center justify-between gap-4):

- Left: Dual pulsing radar indicator (relative flex h-3 w-3 with animate-ping absolute inline-flex h-full w-full rounded-full bg-red-400 opacity-75 + relative inline-flex rounded-full h-3 w-3 bg-red-600), badge CRITICAL RPN 940 • APH CH-VII VIOLATION, and headline: **"Campus Land Deed Survey Alteration & Fire NOC Date Splicing Detected — Apex Institute of Technology & Management (PID: 1-9482710392) — 98.4% TruFor Confidence — Standing Hearing Committee & Physical EVC Recommended"**.
- Right: Two action buttons: View Forensic Evidence (navigates to /applications/AICTE-2026-NW-0842 or opens interactive forensic comparison modal) and Generate Show-Cause / Deficiency Notice (opens pre-populated statutory notice drawer with Sonner toast confirmation).

3. FULL-STACK REPOSITORY ARCHITECTURE & TECH STACK

3.1 Stack Specifications

- **Frontend:** Next.js 16 (App Router), React 19, TypeScript, Tailwind CSS v4, Lucide React icons, sonner for regulatory action notifications, next-themes for seamless Light/Dark mode parity.
- **Backend Service (/backend + Next.js API Route Bridge):** Python FastAPI service (backend/main.py) utilizing PyMuPDF (fitz), OpenCV (cv2), NumPy, Pillow, DuckDB / SQLite analytical state store, and ONNX Runtime inference wrappers for ELA, DCT grid analysis, Shoelace polygon area calculation, and APH compliance scoring. Mirror all FastAPI endpoints inside Next.js Route Handlers (app/api/...) with deterministic algorithmic engines so the prototype runs with zero external dependency failures in any preview environment.

3.2 Application File & Module Manifest

File Path	Architectural Role & Implementation Scope
app/layout.tsx	Loads Google Fonts, wraps application in ThemeProvider, LanguageProvider, FilterProvider, persistent AppShell, and global CommandPalette (⌘K).
app/globals.css	Defines SIH26246 Liquid Glass CSS

	variables, atmospheric radial meshes, 2px severity top rules, and micro-interaction transitions.
app/page.tsx	Entry route redirecting to or directly rendering /overview.
app/overview/page.tsx	Executive Scrutiny Dashboard with the Unified 6-Column KPI Grid, Priority Intervention Banner, Regional Scrutiny Matrix, and High-Risk Queue.
app/applications/page.tsx	Searchable Institutional Dossier Registry with multi-facet filter pills, 5-component ICI spark-bars, and CSV/JSON export.
app/applications/[id]/page.tsx	Dual-Pane Forensic Scrutiny Workbench pairing the interactive multi-spectral document/ELA canvas (left) with extracted entities and APH rules (right).
app/forensics/page.tsx	Digital Forensics Studio with live local file upload ELA/Noiseprint++ sandbox, ResAware vs. InstanceNorm ablation inspector, and ArcFace seal verification.
app/infrastructure/page.tsx	Computer Vision Floor Plan & Campus Audit Studio featuring Mask2Former SVG room polygons, Shoelace area calibrator, and EXIF GPS plot triangulation.
app/authority-bridge/page.tsx	Live National DPI Handshake Monitor for DigiLocker Pull URI XML, EPFO Shram Suvidha ECR 2.0, MCA21, and State e-NOC QR signature verification.
app/deficiencies/page.tsx	Bilingual (EN / हिन्दी) Statutory Deficiency & Show-Cause Notice generator with 15-day EVC rectification countdown tracker.

app/audit-trail/page.tsx	Immutable SHA-256 chained scrutiny ledger logging every AI inference, DPI handshake, and officer sign-off.
app/api/applications/route.ts	Next.js Route Handler serving filtered institutional dossiers and aggregated regional telemetry.
app/api/forensics/analyze/route.ts	Server-side ELA, DCT grid phase, and PDF metadata stream analyzer.
app/api/compliance/calculate/route.ts	Deterministic APH 2024–27 FSI/FAR, Built-up Area (TB_{req}), SFR, Cadre Ratio, ICI, and RPN calculator endpoint.
components/shell/LeftGlassRail.tsx	240px fixed liquid-glass navigation rail with AICTE Emblem, grouped routes, and live anomaly count badges.
components/shell/FloatingTopHeader.tsx	Dynamic breadcrumb, Pipeline Run timestamp, live Authority Bridge pulse pill, ⌘K trigger, EN | हिन्दी switcher, and Theme toggle.
components/shell/GlobalFilterBar.tsx	Contextual glass dropdown bar controlling Academic Year, AICTE Region, Institution Type, Application Category, and Scrutiny Status.
components/workbench/ForensicDocumentCanvas.tsx	Interactive HTML5 Canvas and SVG overlay engine supporting RGB, LayoutLMv3 Bounding Boxes, TruFor Noiseprint++ Heatmap, and ELA sliders.
components/workbench/BlueprintVectorViewer.tsx	Interactive CAD floor plan polygon inspector with graphic scale bar calibration and real-time Shoelace m^2 computation.

lib/mock-data.ts	Complete TypeScript interfaces and 6 deeply populated institutional dossiers across all AICTE regions.
lib/aph-engine.ts	Pure mathematical functions implementing APH 2024–27 Chapter VII infrastructure formulas, SFR/Cadre ratios, ICI gate logic, and RPN scoring.
lib/i18n.ts	100% complete English and formal Hindi (hi-IN) regulatory translation dictionary.
backend/main.py	Companion Python FastAPI service implementing PyMuPDF XRef forensics, OpenCV ELA/Noiseprint++, Shoelace CV, and DuckDB caching.

4. DETAILED VIEW & ROUTE SPECIFICATIONS

4.1 Multi-Layer Application Shell (Persistent Across All Routes)

1. **Left Glass Navigation Rail (240px width on desktop, collapsible drawer on mobile):**
   - Header: Official AICTE Emblem / Regulatory Seal SVG, title AICTE SCRUTINY, subtitle REGULATION BUREAU • DPI v4.2.
   - Grouped Navigation Links with active liquid-glass pill highlight and live anomaly count badges:
     - *CORE INTELLIGENCE:* Executive Overview (/overview), Institutional Registry (/applications - badge 4,820), Scrutiny Workbench (/applications/AICTE-2026-NW-0842 - badge LIVE).
     - *FORENSICS & VISION:* Document Forensics Studio (/forensics - badge 148), Blueprint & Campus CV (/infrastructure - badge 64).
     - *GOVERNANCE & DPI:* Authority Bridge (DPI) (/authority-bridge - badge 98.4%), Deficiency & EVC Notices (/deficiencies), SHA-256 Audit Ledger (/audit-trail).
2. **Floating Top Header:**
   - Dynamic route breadcrumb, Pipeline Run: 4 mins ago, live green pulsing pill DPI Bridge: 6/6 Authorities Online, ⌘K Search PID / UAN / Rule, Language Switcher (EN | हिन्दी), and Light/Dark Mode toggle.
3. **Global Contextual Filter Bar:**
   - Interactive glass dropdowns controlling global state (FilterContext):
     - Academic Year: 2026-27 (Current), 2025-26, 2024-25
     - AICTE Region: All Regions (8), North-Western (Chandigarh/Delhi), Southern (Chennai),

Western (Mumbai), South-Western (Bengaluru), Eastern (Kolkata), Central (Bhopal), Northern (Kanpur), South-Central (Hyderabad)
    - Institution Type: All Types, Private Self-Financed, Government / Aided, Deemed-to-be University, Autonomous
    - Application Category: All Categories, Extension of Approval (EoA), Increase in Intake / New Course, New Technical Institution, Change of Site / Closure
    - Scrutiny Status: All Statuses, Fatal Fraud Alert, Deficiency Issued, Auto-Pre-Cleared, Manual Review Required

4.2 /overview — Executive Scrutiny Dashboard

- **Row 1:** The Unified 6-Column Metric Grid (specified in Section 2.3).
- **Row 2:** The Priority Intervention / High-Risk Alert Banner (specified in Section 2.4).
- **Row 3 (2-Column Asymmetric Split):**
    - **Left (7 Cols): Regional Scrutiny & Anomaly Density Matrix:** Interactive regional breakdown across all 8 AICTE Regional Committees showing applications filed, average Institutional Compliance Index (ICI), Fatal Fraud count, Rectifiable Deficiencies, and Ghost Faculty detections, with clickable regional filters and visual SVG regional bars.
    - **Right (5 Cols): Subsystem Anomaly Telemetry & Authority Bridge Health:** Live breakdown of top forensic triggers (1. Fire NOC Expiry / Date Splicing, 2. EPFO ECR 2.0 / Ghost Faculty Overlap, 3. Blueprint Lab Carpet Area Inflation, 4. Digital Signature / Stamp Cloning, 5. Land Contiguity / FSI Shortfall) plus real-time latency meters for DigiLocker Pull URI, EPFO Shram Suvidha, MCA21 v3, and State e-District QR verifiers.
- **Row 4: High-Risk Institutional Dossier Queue:** Interactive table of top 8 priority applications sorted by Risk Priority Number (RPN), showing PID, Institution Name, Region, Application Type, ICI Score bar, Top Forensic Flag, Authority Handshake badge, and a direct Launch Workbench → button.

4.3 /applications — Searchable Institutional Dossier Registry

- Full-featured registry table + card toggle supporting instant search by Institution Name, AICTE PID, Trust Name, or Survey Number.
- Displays category badges, 5-component ICI breakdown spark-bars (Infra, Faculty, Legal, Authority, Financial), active forensic flags, and batch export to CSV/JSON.

4.4 /applications/[id] — Dual-Pane Forensic Scrutiny Workbench (Hero Technical View)

Implement a synchronized split-pane workstation:

- **Top Workbench Control Bar:** Institution selector dropdown (allowing instant switching between the 6 rich mock institutions), overall ICI score badge, RPN badge, Document Tab Switcher (1. Fire Safety NOC, 2. Land Registry Sale Deed, 3. Architectural Floor Plan, 4. Faculty Roster & EPFO Annex, 5. Audited Balance Sheet), and Officer Decision Buttons (Approve EoA, Issue 15-Day Deficiency Notice, Escalate to Standing Hearing Committee).

- **Left Pane (55% width): Interactive Multi-Spectral Document & Forensics Viewer:**
  - Renders a realistic, high-resolution visual facsimile of the selected institutional document (complete with official headers, tables, stamps, signatures, QR codes, and text blocks).
  - **Interactive Layer Switcher Toolbar:**
    1. Original RGB Document
    2. PP-StructureV3 + LayoutLMv3 Bounding Boxes (color-coded interactive boxes over extracted entities; clicking a box highlights the corresponding rule row in the right pane).
    3. TruFor Noiseprint++ Anomaly Heatmap (renders a calibrated plasma/inferno heatmap isolating spliced dates or pasted stamps with a Confidence Map toggle showing how pristine areas are suppressed).
    4. Error Level Analysis (ELA) & 8×8 JPEG BAG Grid (interactive slider for JPEG Quality $Q \in [50, 95]$ and γ multiplier).
- **Right Pane (45% width): Evidence-Linked Key-Value Extraction, Authority Handshake & APH Rule Checklist:**
  - **Tab 1: Extracted Entities & DPI Reconciliation:** Side-by-side comparison table showing Field Name, OCR Extracted Value (Uploaded PDF), Confidence, Issuing Authority Value (DigiLocker / EPFO / Land Registry), and Status Badge (VERIFIED, MISMATCH_DETECTED, EXPIRED, etc.). Highlight exact mismatches in red (e.g., Uploaded Fire NOC Expiry: 30-06-2027 vs. State Fire Portal QR Record: 30-06-2024).
  - **Tab 2: APH 2024–27 Deterministic Compliance Matrix:** Live mathematical validation of Land FSI/FAR, Built-up Area ($TB_{\text{req}} = \text{INA} + \text{ADA} + \text{AMA} + \text{ACA}$), Student-Faculty Ratio ($1:20$ UG / $1:15$ PG), and $1:2:6$ Cadre Ratio.
  - **Tab 3: Ghost Faculty Deduplication Inspector:** Lists flagged faculty members with side-by-side portrait similarity (ArcFace Cosine Similarity: 0.942), UAN number, primary college vs. concurrent college match in Qdrant, and EPFO ECR 2.0 overlap dates.

4.5 /forensics — Dedicated Digital Forensics Studio

- **Interactive Live Upload & Sandbox:** Allow the user to either select any dossier document OR upload any local image/PDF from their computer to run a live client-side HTML5 Canvas **Error Level Analysis (ELA)**, **Luminance Gradient Analysis**, and **Byte/EXIF Metadata Inspection** in real time.
- **Noiseprint++ Bottleneck & ResAware Architecture Inspector:** Interactive visual comparison demonstrating raw RGB vs. InstanceNorm2d (collapsed signal, AUC 0.554) vs. ResAware + StatNoise Fusion (preserved signal, AUC 0.747) on spliced dates and stamps.
- **Siamese ArcFace Seal & Signature Verification Matrix:** Side-by-side comparison of submitted wet-ink signatures and circular institutional seals against the AICTE Master Specimen Repository, displaying polar coordinate seal unwrapping, cosine similarity $\cos(\theta)$, and SSIM clone detection.
- **PDF XRef & Font Morphology Inspector:** Displays raw PDF /Producer tags, incremental save revisions (%%EOF trailers), and character-level stroke-width / baseline offset graphs

exposing spliced digits.

4.6 /infrastructure — Computer Vision Floor Plan & Campus Audit Studio

- **Interactive Blueprint Vectorizer (Mask2Former + RMBG-1.4 Wall Segmentation):**
  - Renders a multi-room engineering college architectural blueprint with toggleable COCO instance polygon overlays for Classrooms, AICTE Laboratories, Drawing Hall, Admin/Faculty Offices, Amenities/Toilets, Corridors/Circulation, and Structural Walls.
  - **Interactive Scale Bar Calibrator:** User can adjust or click the blueprint's graphic scale bar (10.0m = 240px, $s = 0.04167\ \text{m/px}$) and watch the **Shoelace Formula** recalculate the exact geometric carpet area (m^2) of every room polygon in real time.
  - **Cross-Modal Area Discrepancy Table:** Compares Declared Blueprint Text Area, Portal Claimed Area, CV Shoelace Geometric Area, APH Chapter VII Norm, and Variance % (highlighting an undersized VLSI & Embedded Systems Lab measuring $46.8\ \text{m}^2$ against the $66.0\ \text{m}^2$ APH norm despite being text-labeled as $66.0\ \text{m}^2$).
  - **Interactive What-If FSI/FAR & Intake Simulator:** Sliders for UG/PG Intake, Allied Branch Lab Sharing ($0–30\%$), and Municipal FSI/FAR ($\Phi_{\text{loc}} \in [1.0, 4.0]$) dynamically recomputing TB_{req} and L_{req}.
- **Campus Geotagged Photo & Cadastral Triangulation Audit:**
  - Displays uploaded laboratory equipment photos with **RT-DETR / YOLOv8** bounding boxes (CNC Milling Machine - 0.96, Digital Storage Oscilloscope - 0.94, Wheelchair Ramp - 0.91), Perceptual Hash (pHash) cross-campus duplicate checks, and a multi-plot cadastral map verifying the $\leq 3\ \text{plots}$ and $\leq 2.0\ \text{km}$ inter-plot distance rule.

4.7 /authority-bridge — Issuing Authority Integration Monitor

- Live interactive console displaying request/response payloads for:

1. **DigiLocker / NAD Pull URI API:** Shows the exact XML Request (IssuerId-DocType-DocId), x-digilocker-hmac SHA-256 signature verification, and decoded DataContent XML diff.
2. **EPFO Unified Shram Suvidha & ECR 2.0 Verifier:** Queries Establishment PF Code, 12-digit UAN status, Aadhaar Face Auth (FAT) timestamp, 50% Wage Add-Back compliance, and overlapping employer remittances.
3. **State e-District QR Cryptographic Verifier:** Decodes 2D QR payloads, checks ECDSA/RSA signatures against State Public Keys, and flags spoofing domains.

- Includes a "Run Live Handshake Simulation" button that steps through OAuth PKCE token generation, HMAC signing, and payload reconciliation with realistic latency telemetry.

4.8 /deficiencies — Structured Deficiency Letter Generator & Rectification Tracker

- Automatically compiles all active flags for a selected institution into an official, bilingual

(**English / हिन्दी**) **AICTE Scrutiny Committee Deficiency Notice / Standing Hearing Committee Show-Cause Order**, complete with exact APH 2024–27 chapter/clause citations, forensic evidence thumbnails, and a 15-day EVC rectification countdown timer.

- Supports one-click Sign with Officer DSC, Dispatch to Institution Portal, and Print / Export Official PDF.

4.9 /audit-trail — Immutable Scrutiny Log

- Cryptographically chained scrutiny ledger displaying every AI inference event, DPI API handshake, and Regional Officer override with SHA-256 evidence hashes, officer IDs, and timestamps.

5. EXACT MOCK DATASETS & TYPESCRIPT INTERFACES (lib/mock-data.ts)

Populate lib/mock-data.ts with at least **6 complete, deeply detailed institutional applications** spanning diverse regions, application types, and forensic scenarios so every view, filter, and workbench tab has rich, deterministic data immediately on load:typescript export type VerificationState =
| 'VERIFIED' | 'MISMATCH_DETECTED' | 'EXPIRED' | 'VERIFICATION_PENDING' | 'SOURCE_UNAVAILABLE' | 'MANUAL_REVIEW_REQUIRED';
export type SeverityLevel = 'FATAL_FRAUD' | 'MAJOR_DEFICIENCY' | 'MINOR_DEFICIENCY' | 'COMPLIANT';
export interface ForensicAnomaly { id: string; documentId: string; documentTitle: string; category: 'FORGERY_SPLICING' | 'STAMP_SIGNATURE' | 'BLUEPRINT_AREA' | 'GHOST_FACULTY' | 'AUTHORITY_MISMATCH' | 'PDF_METADATA'; severity: SeverityLevel; titleEn: string; titleHi: string; descriptionEn: string; descriptionHi: string; aphClause: string; modelUsed: string; tamperProbability: number; // e.g., 0.984 confidenceScore: number; // e.g., 0.942 rpnScore: number; // Severity * Occurrence * Certainty (1-1000) boundingBox: [number, number, number, number]; // Normalized [x0, y0, x1, y1] in % extractedValue: string; authorityValue: string; verificationState: VerificationState; }
export interface BlueprintRoom { id: string; name: string; category: 'INA' | 'ADA' | 'AMA' | 'ACA'; svgPolygon: string; // SVG points or rect coordinates declaredAreaSqm: number; // Text printed on blueprint computedAreaSqm: number; // Shoelace formula geometric calculation aphNormSqm: number; // APH Chapter VII minimum requirement status: 'COMPLIANT' | 'INFLATED_LABEL' | 'UNDERSIZED'; }
export interface FacultyRecord { id: string; name: string; designation: 'Professor' | 'Associate Professor' | 'Assistant Professor'; department: string; pan: string; uan: string; // 12-digit EPFO UAN aadhaarFatVerified: boolean; epfoEcrStatus: 'ACTIVE_VERIFIED' | 'DUAL_EMPLOYMENT_DETECTED' | 'NO_REMITTANCE_FOUND'; wageRule50Compliant: boolean; arcFaceSimilarityMax: number; // Cosine similarity against national DB matchedInstitutionName?: string; matchedInstitutionPid?: string; }
export interface InstitutionalApplication { id: string; // e.g., "AICTE-2026-NW-0842" pid: string; // e.g., "1-9482710392" nameEn: string; nameHi: string; region: 'North-Western' | 'Southern' | 'Western' | 'South-Western' | 'Eastern' | 'Central' | 'Northern' | 'South-Central'; state: string; city:

string; zoneType: 'Mega/Metro' | 'Urban' | 'Rural'; institutionType: 'Private Self-Financed' | 'Government / Aided' | 'Deemed-to-be University' | 'Autonomous'; applicationCategory: 'Extension of Approval (EoA)' | 'Increase in Intake / New Course' | 'New Technical Institution' | 'Change of Site / Closure'; scrutinyStatus: 'Fatal Fraud Alert' | 'Deficiency Issued' | 'Auto-Pre-Cleared' | 'Manual Review Required'; academicYear: '2026-27' | '2025-26' | '2024-25'; submittedAt: string; iciScore: number; // 0-100 Institutional Compliance Index fatalGateTriggered: boolean; maxRpn: number; subScores: { infra: number; faculty: number; legal: number; authority: number; financial: number; }; landAndBuiltUp: { applicableFsi: number; totalSanctionedStudentsUG: number; totalSanctionedStudentsPG: number; inaRequiredSqm: number; inaAvailableSqm: number; adaRequiredSqm: number; adaAvailableSqm: number; amaRequiredSqm: number; amaAvailableSqm: number; acaRequiredSqm: number; acaAvailableSqm: number; totalBuiltUpRequiredSqm: number; totalBuiltUpAvailableSqm: number; landRequiredAcres: number; landAvailableAcres: number; plotCount: number; maxInterPlotDistanceKm: number; }; facultyMetrics: { requiredTotal: number; availableVerified: number; ghostFlaggedCount: number; cadreRequired: { prof: number; assoc: number; asst: number }; cadreAvailable: { prof: number; assoc: number; asst: number }; sfrUG: string; sfrPG: string; }; anomalies: ForensicAnomaly[]; blueprintRooms: BlueprintRoom[]; facultyRoster: FacultyRecord[]; }

include these 6 flagship institutional profiles in `lib/mock-data.ts`:

1. **`AICTE-2026-NW-0842`** — Apex Institute of Technology & Management (Gurugram, North-Western). **`Fatal Fraud Alert`** ($ICI = 31.4$, Fatal Gate Triggered, $RPN = 940$). Features a doctored Fire Safety NOC expiry year (`2027` spliced over `2024` detected via TruFor Noiseprint++ & 8×8 JPEG BAG misalignment), a Land Sale Deed survey acreage mismatch against DigiLocker Pull URI (`4.80 Acres` claimed vs. `2.15 Acres` in Haryana Jamabandi revenue record), an inflated `VLSI Design Lab` blueprint polygon ($46.8 m^2$ actual Shoelace area vs. $66.0 m^2$ printed text label), and 4 Ghost Faculty members with overlapping EPFO ECR 2.0 remittances at a sister college in Noida.
2. **`AICTE-2026-SO-1104`** — Sri Venkateswara College of Engineering & AI (Chennai, Southern). **`Deficiency Issued`** ($ICI = 76.8$, $RPN = 420$). Zero forgery (TruFor Integrity ≥ 0.992), all DigiLocker and EPFO handshakes `VERIFIED`, but exhibits rectifiable deficiencies: Associate Professor cadre shortfall (8 available vs. 11 required under 1:2:6), pending State Pollution Control Board renewal (`EXPIRED` by 14 days), and Tutorial Room carpet area shortfall of 4.2%.
3. **`AICTE-2026-WE-0319`** — Sahyadri Autonomous Institute of Information Technology (Pune, Western). **`Auto-Pre-Cleared`** ($ICI = 96.4$, $RPN = 24$). Exemplary 3-year EoA candidate under APH 2024–27; 1:18.2 UG SFR, 100% EPFO ECR 2.0 verified faculty, full barrier-free PWD infrastructure verified by RT-DETR, and surplus built-up area under Pune Metro FSI 2.5.

4. **`AICTE-2026-SW-0592`** — Deccan Academy of Robotics & Architecture (Bengaluru, South-Western): **`Fatal Fraud Alert`** ($ICI = 38.0$, Fatal Gate Triggered, $RPN = 880$). Features digital signature and circular seal cloning ($SSIM = 0.998$ across 9 affidavits), PDF metadata showing modification via `Canva / PDFtk` with unembedded `ArialMT` font injection, and Plot 2 located $3.8km$ away from Plot 1 (violating the APH $\leq 2.0km$ rule).
5. **`AICTE-2026-EA-0715`** — Kalinga Regional Institute of Polytechnic & Engineering (Bhubaneswar, Eastern): **`Manual Review Required`** ($ICI = 81.2$, $RPN = 310$). State Land Revenue server returned HTTP 503 (`SOURCE_UNAVAILABLE`), and the 1994 physical Trust Deed scan has low OCR confidence (0.68), routing to Regional Officer manual visual inspection without penalizing the institution.
6. **`AICTE-2026-CE-0928`** — Vindhya Valley Technical Campus (Bhopal, Central): **`Deficiency Issued`** ($ICI = 72.5$, $RPN = 490$). Applying for increase in intake in B.Tech AI & Data Science; laboratory equipment photo pHash/DINOv2 audit detected 2 computer lab photographs reused from the Mechanical CAD lab submission, plus 3 faculty members failing the 50% basic wage add-back rule in EPFO ECR 2.0 filings.

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6. VERIFICATION, POLISH & ACCEPTANCE CRITERIA

1. **Zero Build or Type Errors:** Run `npm run build` / `npx tsc --noEmit` to verify strict TypeScript compliance across all pages and components.
2. **Interactive Filter & Search Reactivity:** Changing any dropdown in the `GlobalFilterBar` (Academic Year, Region, Institution Type, Category, Status) or typing in the `⌘K` Command Palette MUST dynamically filter the KPI counts, regional matrix, and institutional registry.
3. **Genuine Interactive Canvas Tools:** Ensure the `/applications/[id]` Workbench and `/forensics` Studio render interactive SVG/HTML5 Canvas overlays where toggling `Bounding Boxes`, `TruFor Noiseprint++ Heatmap`, and `ELA Recompression` visibly transforms the document inspection canvas and updates the linked right-hand inspector panel.
4. **Responsive & Theme-Complete:** Verify layout integrity from `375px` mobile viewports to `1920px` widescreen regulatory monitors in both Light (`#F4F7FB`) and Dark (`#0B132B`) modes, and verify that switching `EN | हि-IN` updates all UI labels cleanly.

6. Strategic Synthesis and Architectural Conclusions

The technical blueprint developed for Problem Statement 26261 resolves the core tension in regulatory automation: achieving high-throughput, mathematical rigor across thousands of institutional dossiers without sacrificing procedural fairness or forensic explainability[1]. Decoupling fatal fraud from rectifiable deficiencies prevents administrative paralysis during annual approval cycles; by gating the Institutional Compliance Index (ICI) through a calibrated TruFor confidence map ($C_{\mathrm{conf}} \geq 0.80$) and deterministic API Setu and EPFO cryptographic

handshakes, the platform isolates the subset of applications engaging in active document tampering or Ghost Faculty syndicates while providing transparent, automated 15-day rectification checklists for genuine institutions with minor infrastructural or cadre variances[6]. Furthermore, signal-preserving neural forensics overcomes the extreme spatial sparsity of document manipulations. Standard computer vision pipelines fail on regulatory filings because tampered digits occupy less than 1% of the page and spatial normalization layers (InstanceNorm2d) erase sensor noise statistics[18]. Enforcing full-frame native Noiseprint++ extraction prior to cropping, replacing instance normalization with batch-normalized ResAware and StatNoise scalar fusion, and pairing neural heatmaps with deterministic PDF XRef and 8×8 DCT grid forensics provides legally defensible evidence for AICTE Standing Hearing Committees[19]. Finally, closed-loop DPI handshakes eliminate cross-institutional arbitrage: integrating PP-StructureV3 and SLANet table extraction directly with 512-dimensional ArcFace vector indexing and EPFO Shram Suvidha ECR 2.0 payroll verification makes it mathematically impossible for a single faculty member or laboratory photograph to be double-counted across multiple technical campuses in India[7].

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